

Vishesh Sinha

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SUMMARY

Curious AI builder who keeps asking what comes next: how agents will plan and use tools, how small local models can do the work of cloud ones, and how we can measure and trust what a model tells us. I learn by turning ideas and research into working systems, such as agentic LLM workflows, retrieval pipelines for Indic languages and local-first AI that runs on everyday laptops, and then testing them honestly. I have worked on a Hospital ERP module at STPI and built the full backend of a fintech app as a freelancer. I want to work on agentic AI, information retrieval and trustworthy machine learning, where there is always something new to figure out.

EDUCATION

Poornima Institute of Engineering and Technology

2024 – 2028

B.Tech in Artificial Intelligence & Data Science | Current SGPA: 8.8 / 10

EXPERIENCE

Intern, Software Technology Parks of India (STPI)

May 2026 – Jul 2026

Hospital ERP System

- Developed the **Medical Tourism module** of a Hospital ERP system, covering how international patients are registered, managed and tracked inside the hospital management workflow.
- Worked with the team to test the module and integrate it into the wider ERP so it worked with the existing hospital modules.

Freelance Developer, Financial Tracking App

Nov 2025 – Apr 2026

Backend, Architecture and AI Integration

- Designed the overall **architecture** of the project and built the **entire backend** myself, owning the structure of the system from data handling to the APIs the app depends on.
- Integrated the **AI features** of the app by connecting AI models to users' financial data, so the app could return smarter insights instead of plain records.
- Managed the project end to end as a freelancer: requirements, design decisions, implementation and delivery.

PROJECTS

ETRIGAN: Local AI Workbench for Low-Spec Laptops

Python, Ollama, SQLite, WSL, CLI

- Built a CLI-first workbench that makes local LLMs and agents practical on a laptop with only **4 GB VRAM and 7.8 GB RAM**, instead of assuming a cloud service or a high-end GPU.
- Hardware-aware commands: **doctor** checks the GPU, RAM, disk and tooling, **bench** measures speed and memory per model, and **fit** recommends which models and quantizations suit the machine.
- Memory-aware model router, hybrid retrieval (BM25 and vector search) that returns cited answers and refuses when evidence is weak, an offline code sandbox, and a hash-chained audit log that makes tampering detectable.

Voice-Enabled Indic RAG Pipeline

Python, FastAPI, Sentence-Transformers, Sarvam AI, ElevenLabs, faster-whisper

- Built an end-to-end retrieval-augmented generation system for **Hindi and other Indic languages** on the AI4Bharat MSMARCO-XI dataset, served through FastAPI with a web dashboard.
- Fine-tuned the **multilingual-e5-small** embedding model and implemented six chunking strategies, including Devanagari-aware splitting, combined with dense and BM25 retrieval.
- Added pluggable speech services for live voice queries, a **four-tier guardrail system** that refuses low-confidence answers, and an error-recovery layer so the pipeline keeps running when a service fails.

Agentic Fraud Investigation System

TigerGraph, LangGraph, Google Gemini, Model Context Protocol (MCP), Streamlit, Python

- Built an autonomous fraud investigator over a graph of about **590K transactions**, 14K cards and 9.7K device profiles stored in TigerGraph.
- Designed an **8-stage LangGraph workflow** that investigates an alert, gathers evidence, assesses its own uncertainty, explains the findings and updates case memory, then recommends a policy action and drafts a Suspicious Activity Report.
- Exposed graph traversals and GraphRAG as tools through a custom **MCP server**, ran the agent on 20 benchmark cases, and delivered a Streamlit dashboard for analysts.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, Rust, Julia, Bash, SQL, GSQL

Frameworks & Libraries: FastAPI, Uvicorn, LangGraph, LangChain, PyTorch, scikit-learn, NumPy, Pandas, Sentence-Transformers, Streamlit

Databases: PostgreSQL, MySQL, SQLite, TigerGraph

Tools & DevOps: Git, GitHub, CI/CD, Docker, Ollama, Linux / WSL

Concepts: RAG, GraphRAG, Hybrid Retrieval, Agentic Workflows, Model Context Protocol (MCP), LLM Guardrails, Machine Learning, Model Evaluation

CERTIFICATIONS

CS50x: Introduction to Computer Science, Harvard University (edX) — Topics: C, Python, SQL, HTML, CSS, JavaScript, Algorithms, Data Structures

NPTEL: Introduction to Machine Learning